PassFrame tuition

Volume & 3D Shapes

Year 5 Accelerated — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. 30cm³

2. 64cm³

3. C — both A (36cm³) and B (36cm³)

4. 12

5. 8

6. 36cm³

7. B — cylinder

8. 3

9. A, B, C and D (a triangular prism has 5 faces, 9 edges, 6 vertices and 2 triangular faces — not all faces are rectangles)

10. 125cm³

11. 5

12. 32cm³

13. 9

14. 0

15. 27cm³

Standard

1. 3cm

2. 18

3. 2 litres

4. A — 5cm

5. 200cm³

6. 1 (the curved edge where the flat circular face meets the curved surface)

7. 15 litres

8. A, B and D (this net folds into a cube with 6 square faces)

9. 4cm

10. 5

11. B — 27cm³ is greater than A's 24cm³

12. 4cm

13. 3500cm³

14. 70cm³

15. 6

Standard / Hard

1. 15 litres

2. B — height 4cm (240÷60); 4 boxes (960÷240)

3. Edge = 8cm; surface area = 6 × 64 = 384cm²

4. 108 m³; 108,000 litres

5. 6cm

6. A, C and D (edge = 8cm, surface area = 384cm², and it has 8 vertices)

7. 25 litres

8. 12×8×4 = 384cm³

9. 20cm (1000 ÷ 50)

10. 3m³ = 3000 litres; 6 minutes

11. 24 cubes

12. 9cm × 8cm × 2.5cm each

13. Width 6cm, length 12cm (since 12 × 6 × 5 = 360)

14. Container = 8000cm³; each bottle = 1000cm³; 8000 ÷ 1000 = 8 bottles filled, 0 left over

15. √(20×15×12) = √3600 = 60cm³

Homework — Answers

Easy / Standard

1. 40cm³

2. 216cm³

3. C — both A (42cm³) and B (42cm³)

4. 12

5. 8

6. 42cm³

7. B — cone

8. 2

9. A, B and C (a cylinder has 3 faces total: 2 flat circles and 1 curved surface; it has no straight edges or vertices in the usual sense)

10. 343cm³

11. 8

12. 54cm³

13. 6

14. 2 (the two circular edges where the curved surface meets each flat face)

15. 1000cm³

Standard

1. 7cm

2. 20

3. 4.5 litres

4. A — 6cm

5. 144cm³

6. 1

7. 20 litres

8. A, B and D (this net folds into a triangular prism with 5 faces, 2 of them triangular)

9. 4cm

10. 6

11. B — 24cm³ is greater than A's 20cm³

12. 5cm

13. 2200cm³

14. 96cm³

15. 9

Standard / Advanced

1. Edge = 6cm, so volume = 6×6×6 = 216cm³

2. B — 80cm³ (height = 8cm, so 5×2×8=80)

3. Cuboid's volume = 216cm³, so the cube's edge = 6cm (since 6×6×6=216)

4. A, B and C (width = 5cm, height = 5cm, so volume = 20×5×5 = 500cm³)

5. 6cm

6. Missing length = 7cm; largest face area = 42cm² (7cm × 6cm)

7. 1331cm³; side = 11cm

8. √(8×6×12) = √576 = 24cm³

9. 900 ÷ 3 × 2 = 600 ml

10. 27 litres

11. Edge = 9cm; area = 6 × 81 = 486cm²

12. 192,000cm³ = 192 litres; 24 minutes

13. 24 blocks

14. 0.48 m³ = 480 litres

15. 32 litres each